

# Subset & Subsequences

"Motivation is what gets you started. Habit is what keeps you going."

~ Jim Ryun



BRIGHT  
DROPS  
.com

Good  
Evening



## Today's content

01. Modular Arithmetics
02. Basic AND/OR principle
03. Intro to subset vs Subsequence
04. Subsequence with a given sum  $\rightarrow$  ?

## Modulus operator (%)

$a \% m \rightarrow$  Remainder of  $a$  when divided by  $m$

$$10 \% 4 = 2 = 10 - \{\text{largest mul of } 4 \leq 10\} 8 = 2$$

$$13 \% 5 = 3 = 13 - \{\text{largest mul of } 5 \leq 13 = 10\} = 3$$

$$\text{Dividend} = \text{divisor} * \text{quo} + \text{rem}$$

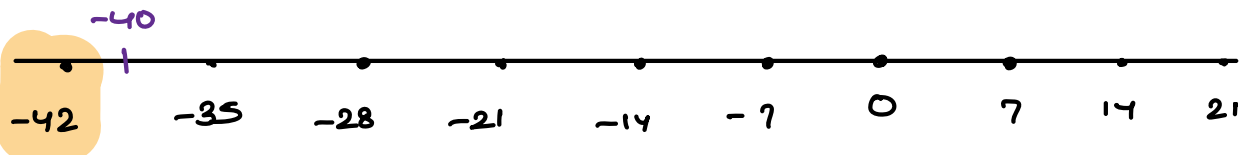
$$\text{Rem} = \text{Dividend} - (\text{divisor} * \text{quotient})$$

$\{\text{largest multiple of divisor} \\ \leq \text{dividend}\}$

$$150 \% 11 = 150 - \{\text{largest mul of } 11 \leq 150 = 143\} = 7$$

$$20 \% 7 = 20 - \{\text{largest mul of } 7 \leq 20 = 14\} = 6$$

$$\begin{aligned} -40 \% 7 &= -40 - \{\text{largest mul of } 7 \leq -40 = -42\} \\ &= -40 - (-42) = -40 + 42 = 2 \end{aligned}$$



$$-60 \% 9 = -60 - (-63) = -60 + 63 = 3$$

$$\begin{aligned} -40 \% 9 &= -40 - (\text{largest mul of } 9 \leq -40) \\ &= -40 - (-45) = -40 + 45 = 5 \end{aligned}$$

$$x \% 4 = 0, 1, 2, 3$$

$$x \% 6 = 0, 1, 2, 3, 4, 5$$

Obs

$$x \% M = \left\{ \overset{\text{min}}{0} \quad \overset{\text{max}}{m-1} \right\}$$

Issue with Java/C++/C#/JS

|                 | Java                  | (Python)<br>Expected |
|-----------------|-----------------------|----------------------|
| print (-40 % 7) | -5 $\xrightarrow{+7}$ | 2                    |
| print (-60 % 9) | -6 $\xrightarrow{+9}$ | 3                    |
| print (-40 % 9) | -4 $\xrightarrow{+9}$ | 5                    |

In Java, for -ve values. we want to get correct ans

| a   | m | code                                                                                                                | Exp |
|-----|---|---------------------------------------------------------------------------------------------------------------------|-----|
| -63 | 9 | <div><div>a &gt; 0</div><div>return a % m</div><div>else {</div><div>return (a % m + m) % m</div><div>}</div></div> | 0   |
| -40 | 9 | 5                                                                                                                   | 5   |

$$m = 10^9 + 7 ? \rightarrow \text{Hashing}$$

$a \% b$  in Java  $\rightarrow \frac{a}{b} = \text{integer division}$

$$\begin{aligned} 100 \% 7 &= 100 - (7 \times y) = \\ &= 100 - (7 \times 14) = 2 \end{aligned}$$

$\rightarrow y = \frac{100}{7} = 14$

$$\begin{aligned} -40 \% 7 &= -40 - (7 \times y) \\ &= -40 - (-7 \times 5) \\ &= -5 \end{aligned}$$

$\rightarrow y = \frac{-40}{7} = -5$

$a \% b$  in Python

$$\frac{a}{b} = \text{floor}(a/b)$$

$$\frac{6}{4} = \text{floor}(1.5) = 1$$

$$\frac{7}{3} = \text{floor}(2.33...) = 2$$

$$\begin{aligned} 100 \% 7 &= 100 - (7 \times y) \\ &= 100 - 98 \\ &= 2 \end{aligned}$$

$\rightarrow y = \text{floor}(\frac{100}{7}) = 14 \dots = 14$

$$\begin{aligned} -40 \% 7 &= -40 - (7 \times y) \\ &= -40 - (-7 \times 6) \\ &= -40 + 42 = 2 \end{aligned}$$

$\rightarrow y = \text{floor}(\frac{-40}{7}) = (-5 \dots) = -6$

Why M?  $\rightarrow$  Limit our output data in given range  
 $\rightarrow$  To prevent overflow

$$\left. \begin{array}{c} -\infty \\ \infty \end{array} \right\} \% M = \text{Expected output} = \{0 \text{ to } m-1\}$$

Modular arithmetics  $\rightarrow$  ( $\%$ ) with (+, -, \*, /)

01.  $(a+b) \% m = (a \% m + b \% m) \% m$

$$\frac{a}{13} \quad \frac{b}{9} \quad \frac{m}{5}$$

$$\begin{aligned} (13+9) \% 5 &\leftrightarrow (13 \% 5 + 9 \% 5) \% 5 \\ 22 \% 5 &= (3 + 4) \% 5 \\ = 2 &= 7 \% 5 = 2 \end{aligned}$$

$$(0-m-1) * (0-m-1)$$

02.  $(a * b) \% m = (a \% m * b \% m) \% m$

$$\frac{a}{13} \quad \frac{b}{9} \quad \frac{m}{5}$$

$$\begin{aligned} (13 * 9) \% 5 &= (13 \% 5 * 9 \% 5) \% 5 \\ = 117 \% 5 = 2 &= (3 * 4) \% 5 = 12 \% 5 = 2 \end{aligned}$$

$$a = 10^9 \quad m = 10 \quad = \underbrace{(10^9 \% 10 * (10^9 \% 10))}_{(9 * 9)} + 1010$$

$$b = 10^9$$

$$\underline{81} \% 10 \Rightarrow \boxed{1}$$

03.  $(a-b) \% m = (a \% m - b \% m + m) \% m$

$$\frac{a}{13} \quad \frac{b}{9} \quad \frac{m}{5}$$

$$(13-9) \% 5 = (13 \% 5 - 9 \% 5) \% 5$$

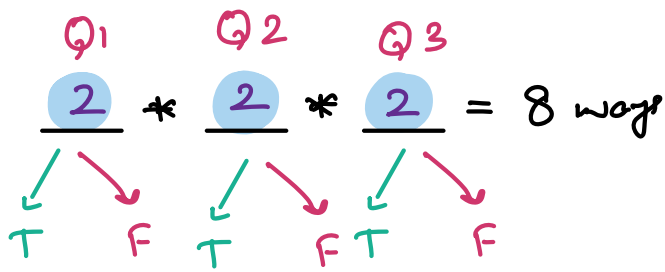
$$4 \% 5 = (3 - 4) \% 5$$

$$= 4 = -1 \% 5$$

$$= -1$$

## AND/OR PRINCIPLE

01. Given 3 True/False questions, every question has to be answered. In how many ways can you answer all the questions.



You need to answer Q<sub>1</sub> **and**

you need to answer Q<sub>2</sub>

**and**

you also need to answer Q<sub>3</sub>

- |     |   |   |   |
|-----|---|---|---|
| 01. | F | F | F |
| 02. | F | F | T |
| 03. | F | T | F |
| 04. | F | T | T |
| 05. | T | F | F |
| 06. | T | F | T |
| 07. | T | T | F |
| 08. | T | T | T |
- } 8 ways

AND  $\rightarrow *$

OR  $\rightarrow +$

Q2 Given 5 girls & 4 boys. How many different pairs can we form?

Note :- pair  $\rightarrow$  1 Girl + 1 Boy

Boys

B<sub>1</sub>

B<sub>2</sub>

B<sub>3</sub>

B<sub>4</sub>

Girls

G<sub>1</sub>

G<sub>2</sub>

G<sub>3</sub>

G<sub>4</sub>

G<sub>5</sub>

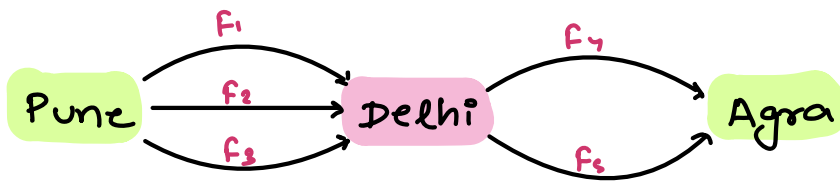
Idea  $\rightarrow$  Pick 1 Boy out of 4

**and**

pick 1 Girl out of 5

$$\text{Ans} = 4 * 5 = 20$$

Eg 3



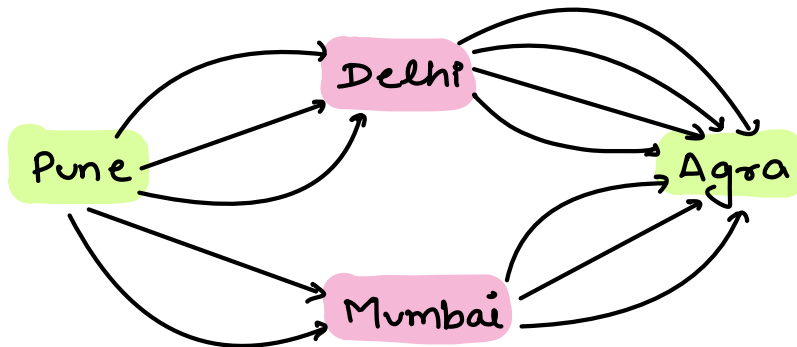
No. of ways to reach Agra from Pune via Delhi

$$\text{Ways} = \text{Pune} \rightarrow \text{Delhi} \text{ AND } \text{Delhi} \rightarrow \text{Agra}$$

$$3 * 2 = 6$$

$F_1 F_4$      $F_2 F_4$      $F_3 F_4$   
 $F_1 F_5$      $F_2 F_5$      $F_3 F_5$

Eg 4



No. of ways to reach Agra from pune?

WAYS

$$\underbrace{P \rightarrow D}_{3} \text{ AND } \underbrace{D \rightarrow A}_{4} \quad \text{OR} \quad \underbrace{P \rightarrow M}_{2} \text{ AND } \underbrace{M \rightarrow A}_{3}$$

$$= 12 + 6 = 18 \text{ ways}$$



# Subarrays vs Subseq vs Subsets

Subarrays → Continuous part of an array

Subsequence → Sequence obtained by deleting none or more ele from array

01. Data should be continuous w.r.t indices (Order should be maintained)

02. Empty sequence is also a subseq.

arr = { 3 2 1 9 6 8 }  
          0 1 2 3 4 5

Subarr

Eg:- { 1 9 6 } → Yes

Eg:- { 3 9 8 } → No

Eg:- { 9 6 3 } → No

Eg:- { 2 } → Yes

Eg:- { } → No

Eg:- { 3, 2, 1, 9, 6, 8 } → Yes

Subseq

{ \* \* 1 9 6 \* }  
Yes

{ 3 \* \* 9 \* 8 }  
Yes

No

Yes

Yes

Yes

Obs  $\rightarrow$  All subarrays are subseq

All subseq are not subarrays

\* —  $\alpha$  — + —  $\alpha$  — +

arr = {3, 1, 8}  $\xrightarrow{\text{sort}}$  {1, 3, 8}

All subseq

All subseq

|           |                                                          |           | <u>sum</u> |
|-----------|----------------------------------------------------------|-----------|------------|
| { }       | $\longleftrightarrow$                                    | { }       | 0          |
| {3}       | $\longleftrightarrow$                                    | {3}       | 3          |
| {1}       | $\longleftrightarrow$                                    | {1}       | 1          |
| {8}       | $\longleftrightarrow$                                    | {8}       | 8          |
| {3, 1}    | $\xrightarrow[\text{same}]{\text{Order diff, but data}}$ | {1, 3}    | 4          |
| {1, 8}    | $\longleftrightarrow$                                    | {1, 8}    | 9          |
| {3, 8}    | $\longleftrightarrow$                                    | {3, 8}    | 11         |
| {3, 1, 8} | $\xrightarrow[\text{same}]{\text{Order diff, but data}}$ | {1, 3, 8} | 12         |

Observation → If we sort the arr

↳ The order may differ  
but the data remains same

|               |   |                                |
|---------------|---|--------------------------------|
| Sum of subseq | } | Are not affected<br>by sorting |
| XOR of subseq |   |                                |
| max of subseq |   |                                |
| min of subseq |   |                                |

Conclusion → Prop that depends on the element  
& not the indices are going to  
remain same ↓ even after the sorting  
in subsequences

10:30 → 10:37 pm

Subsets → Collections of elements from the array /  
Delete none or more ele from array

- Order can change w.r.t indices
- Continuity is not important.

arr[] = { 4 -1 3 9 2 6 14 }

|            | subarray | subseq | subset |
|------------|----------|--------|--------|
| { 4 -1 3 } | Yes      | Yes    | Yes    |
| { -1 3 6 } | No       | Yes    | Yes    |
| { 9 4 14 } | No       | No     | Yes    |
| { }        | No       | Yes    | Yes    |

Obs 1 → Every subseq is a subset

→ Every subset is not a subseq

→ Every subarr is a subseq & it is also a subset

• Arr = { 1, 2, 3, 4, 5 }

Subsets → { 1, 4, 3, 5 } = { 1, 3, 4, 5 }

{ 1, 5, 3, 4 }

{ 1, 5, 4, 3 }

{ 3, 1, 4, 5 }

{ 3, 4, 5, 1 }

⋮  
 All these subsets  
 are same

Statement = No. of subsets = No. of subsequences

arr = [1, 2, 3]

$\text{Subset} = \{1, 2, 3\} \xrightarrow{\text{subseq}} = \{1, \cancel{2}, \cancel{3}\}$   
 $= \{2, \cancel{3}, \cancel{1}\}$   
 $= \{2, \cancel{1}, \cancel{3}\}$   
 $= \{3, \cancel{1}, \cancel{2}\}$   
 $\vdots$

Count of subseq & subsets

$\text{arr}[3] = \{6 \quad 7 \quad 8\}$   
  
 $2 * 2 * 2 = 8$

$\text{arr}[n] = \{a_1 \quad a_2 \quad a_3 \quad a_4 \quad \dots \quad a_n\}$

No. of subsets =  $2 * 2 * 2 * 2 * \dots * 2 = 2^n / 1 \leq n$

No. of subseq

Q - Given an  $ar[N]$ , check if there exist a subsequence with  $sum = k$

### Constraints

$$1 \leq N \leq 20$$

$$1 \leq ar[i] \leq 10^6$$

Eg:  $ar[5] = \{ \underset{0}{2} \underset{1}{5} \underset{2}{3} \underset{3}{11} \underset{4}{7} \}$

$$K = 20 \rightarrow \text{True } \{2, 11, 7\}$$

$$K = 8 \rightarrow \text{True } \{5, 3\}$$

$$K = 9 \rightarrow \text{True}$$

$$K = 19 \rightarrow \text{True}$$

$$K = 50 \rightarrow \text{False}$$

Idea  $\rightarrow$  For all subseq, get the sum & compare it with  $K$

if ( $sum == k$ ) return True

else return False.

Generate of subseq

int C arr = { 7 5 3 }

select an ele : 1  
delete an ele : 0

Decimal

|   |   |   |   |               |
|---|---|---|---|---------------|
| 0 | 0 | 0 | 0 | = { }         |
| 1 | 0 | 0 | 1 | = { 3 }       |
| 2 | 0 | 1 | 0 | = { 5 }       |
| 3 | 0 | 1 | 1 | = { 5, 3 }    |
| 4 | 1 | 0 | 0 | = { 7 }       |
| 5 | 1 | 0 | 1 | = { 7, 3 }    |
| 6 | 1 | 1 | 0 | = { 7, 5 }    |
| 7 | 1 | 1 | 1 | = { 7, 5, 3 } |

Final Idea → Get the decimal no. → { 0 to  $2^n - 1$  }  
range

↓  
generate the binary for particular decimal no.

↓  
take the ele of that particular id

## Pseudocode →

for ( $i=0$  ;  $i < 2^n$  ;  $i++$ ) {  $i \rightarrow$  represent the number

$sum=0$  ,  $num=i$

→ for ( $j=0$  ;  $j < n$  ;  $j++$ ) { → To find the corresponding subseq

if ( $(num \& 1 == 1)$ )

|  $sum = sum + \underline{arr[n-1-j]}$

|  $num = num \gg 1$  ;

}

if ( $sum == k$ ) return true

}

return false

$TC = O(2^n * n)$

$SC = O(1)$



# Doubts

## Batches

### Beginner

- Freshers → coming out of college
- Sales
- Writing code for the first time
- Difficulty level ↓

### Intermediate

- Experienced people
- Writing codes & sign
- Difficulty level ↑

Q →

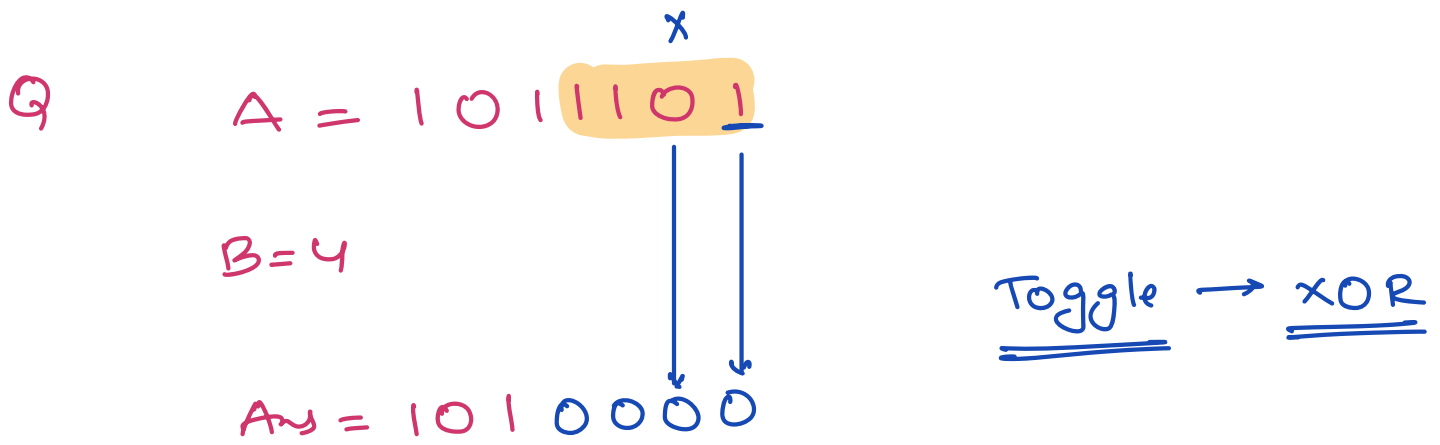
$$\begin{array}{r} \textcircled{A} \\ \hline 11 \\ \downarrow \\ 1 \end{array} \quad \& \quad \begin{array}{r} B \\ \hline 1 \\ \downarrow \\ 2 \end{array}$$

num = 0

$$\begin{array}{cccccccccccc} \text{num} = & 0 & 0 & 0 & 0 & 0 & \dots & 0 & 0 & 0 & 0 \\ & 0 & 0 & 0 & 0 & 0 & \dots & 0 & 1 & 1 & 0 \end{array}$$

$$\text{num} = \text{num} \mid (1 \ll A)$$

$$\text{num} = \text{num} \mid (1 \ll B)$$

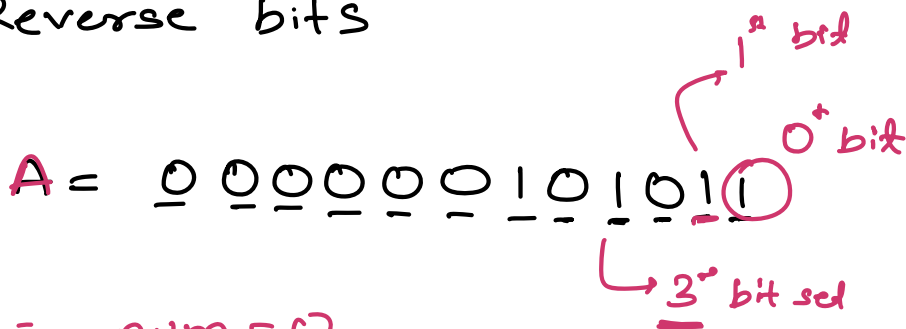


Write a loop that runs for B times

if (bit is set or not)

↳ Take the XOR

\* Reverse bits



Idea = num = 0

write a for loop for all 32 bits (

if (i<sup>th</sup> bit is set in A)

↳ num = num | (1 << (31 - i))

}



$$31-1$$

$$31-3$$

|   | 0  | 1  | 2  | 3  | 4  |
|---|----|----|----|----|----|
| 0 | 1  | 2  | 3  | 4  | 5  |
| 1 | 6  | 7  | 8  | 9  | 10 |
| 2 | 11 | 12 | 13 | 14 | 15 |
| 3 | 16 | 17 | 18 | 19 | 20 |
| 4 | 21 | 22 | 23 | 24 | 25 |

$$n = 5$$

$$\lfloor n-1 \rfloor = 4$$

$$n = 5 - 2 = 3$$

$$2$$